

Multivariate Analysis of Riot Games Match Data: PCA, Clustering and Discriminant Analysis

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Abstract

This report presents a multivariate analysis pipeline applied to Riot Games match data. Steps include data collection from the Riot API, data cleaning and sampling, dimensionality reduction (PCA), clustering (K-Means), and discriminant analysis (LDA). The analysis is applied to two distinct modelling targets: player rank classification and match outcome prediction. We emphasize visual interpretation of results and business impact for analytics products competing with existing platforms.

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1 Introduction and Context

This study uses a large dataset of League of Legends matches collected via the Riot Games API. The raw ingestion pipeline yields JSON objects per match and per participant. The project’s objective is to extract robust, interpretable player and match-level insights that can be operationalized in dashboards and prediction services.

Key goals:

- Demonstrate a principled pipeline from raw API JSON to compact datasets suitable for modeling.
- Produce interpretable visualizations (PCA biplots, radar charts, ridge plots) and rigorous clustering.
- Validate predictive potential using classification/regression models and LDA.
- Provide actionable business recommendations for productization.

2 Data Collection and Dataset Construction

2.1 Riot API Ingestion and JSON Architecture

Data was collected continuously over several weeks using an authenticated Riot API key. The collector queries the *Match-V5* and *Timeline* endpoints, saving each match as a highly nested JSON record.

At the end of the collection phase, the raw data amounted to a massive **11.4 GB** of JSON files. Because League of Legends is an incredibly complex game, each individual JSON match file contains roughly **3,500 distinct features** when completely flattened. This includes:

- **Match metadata:** gameId, mapId, gameDuration, queueId, timestamp.
- **Participant stats:** kills, deaths, assists, goldEarned, damageDealtToChampions, visionScore, wardsPlaced, etc.
- **Timeline events:** objective timestamps (dragons, barons), item purchases, spell casts, and precise player coordinates recorded every minute.

2.2 Creation of 4 Specialized Datasets

Handling 3,500 features simultaneously triggers the “Curse of Dimensionality” and causes severe multicollinearity. From this 11.4 GB raw set, we applied domain-driven feature engineering to select exactly what we needed to create **four distinct, compact datasets**, each tailored for a specific machine learning model:

1. Rank Classification Dataset:

- *Goal:* Predict a player’s ELO tier based on historical performance.

- *Key Features:* avg_kda, avg_cs_per_min, avg_gold_per_min, avg_damage_per_min, avg_vision_score, win_rate, soloKills, and turretPlates.

2. Match Outcome Dataset (Early & Full Game):

- *Goal:* Predict whether a team will win or lose.
- *Key Features:* Differential metrics such as gold_diff_at_15, xp_diff_at_15, first_blood_secured, first_dragon_secured, team_total_cc_duration, and damage_to_objectives.

3. Smurf Anomaly Detection Dataset:

- *Goal:* Detect smurfs (highly skilled players on low-rank accounts) using unsupervised anomaly detection.
- *Key Features:* win_rate_recent (sudden spikes), kda_consistency (abnormal variance), matchmaking_time, and champion_pool_size.

4. Progression Regressor Dataset:

- *Goal:* Predict a player’s future win-rate trajectory (Δ win-rate).
- *Key Features:* recent_form_10_matches, recent_form_30_matches, gold_std, and role_focus_pct.

2.3 Match Outcome Dataset: Data Construction and Iterative Improvement

Constructing a well-balanced and structurally sound dataset for match outcome prediction required two design iterations, each addressing a specific methodological flaw discovered in the previous version.

Version 1 — Naive Balancing Across Different Matches. In the first iteration, we assembled the dataset by sampling independently from winning and losing matches:

- 50% of rows drawn from winning team records across M_{win} .
- 50% drawn from losing team records across a **different** set M_{loss} , with $M_{\text{win}} \cap M_{\text{loss}} = \emptyset$.

While this achieved class balance, it introduced a critical structural flaw: the two opposing teams of the same match were never jointly represented. Differential features such as gold_diff_at_15 were computed in isolation, losing the zero-sum nature of the game. The model learned absolute performance levels rather than relative competitive advantage.

Version 2 — Paired Team Records from the Same Match (Improved). In the second iteration, for every match m_i included in the dataset, **both teams** are included as two separate rows:

Row	Match ID	Team	Outcome
$2i - 1$	m_i	Blue (winner)	1
$2i$	m_i	Red (loser)	0

This design guarantees:

- **Structural symmetry:** differential features are perfectly anti-correlated across the two rows of the same match, faithfully reflecting the zero-sum game mechanic.
- **Natural class balance:** every match contributes exactly one winning row and one losing row.
- **Richer signal:** the model learns *relative* advantage patterns rather than absolute stat thresholds.
- **Reduced selection bias:** both outcomes of each match are present, eliminating systematic sampling distortions.

This structural improvement led to a measurable increase in model generalization and distinguishes our pipeline from naive win/loss sampling approaches.

2.4 Preprocessing and Sampling

Raw ingestion yields a dataset of $\sim 350,000$ matches. To enable exploratory analysis and clustering without exceeding RAM, we sample 10,000 matches using stratified sampling by queue type and patch version. A properly stratified sample of 10k provides stable estimates for cluster centroids and PCA loadings while remaining computationally tractable.

Outlier handling:

- IQR method ($1.5 \times \text{IQR}$) applied to strongly skewed metrics (KDA, damage) to clip extreme outliers before scaling, as K-Means is sensitive to extremes.
- Missing values imputed with median for numeric features; a special category assigned for categorical fields (e.g., role = unknown).

3 Exploratory Data Analysis

3.1 Feature Correlations and Multicollinearity

Before PCA, correlation matrices highlight multicollinearity among the rank classification features (e.g., gold $\leftrightarrow$ CS, kills $\leftrightarrow$ damage). PCA is chosen to summarize correlated variation and produce orthogonal axes for clustering.

Matrice de corrélation (Identifier la colinéarité)

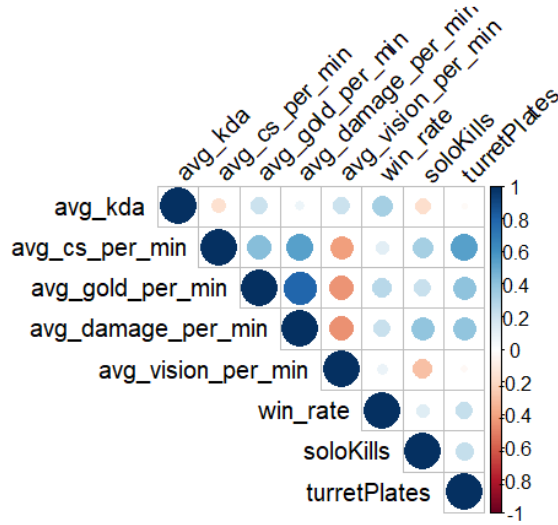


Figure 1: Correlation matrix of rank classification features. Strong positive associations are visible between economy-related metrics (CS/min, gold/min, damage/min), confirming multicollinearity that motivates PCA. Inverse relations — notably between death-related metrics and performance indicators — reveal opposing behavioural dimensions within the feature space.

The correlation matrix (Figure 1) confirms that raw features cannot be fed directly into a distance-based model: highly correlated variables would implicitly over-weight a single underlying dimension (economy output). PCA resolves this by constructing orthogonal components that separate these co-moving variables into distinct axes.

3.2 CS per Minute Distributions by Skill Tier

As a preliminary univariate sanity check, we examine the distribution of `avg_cs_per_min` across the three broad ELO groups derived in Section 9.2. This single feature alone illustrates the value of grouping: the distributional overlap between adjacent individual ranks is substantial, whereas the three merged tiers show a cleaner, more interpretable separation.

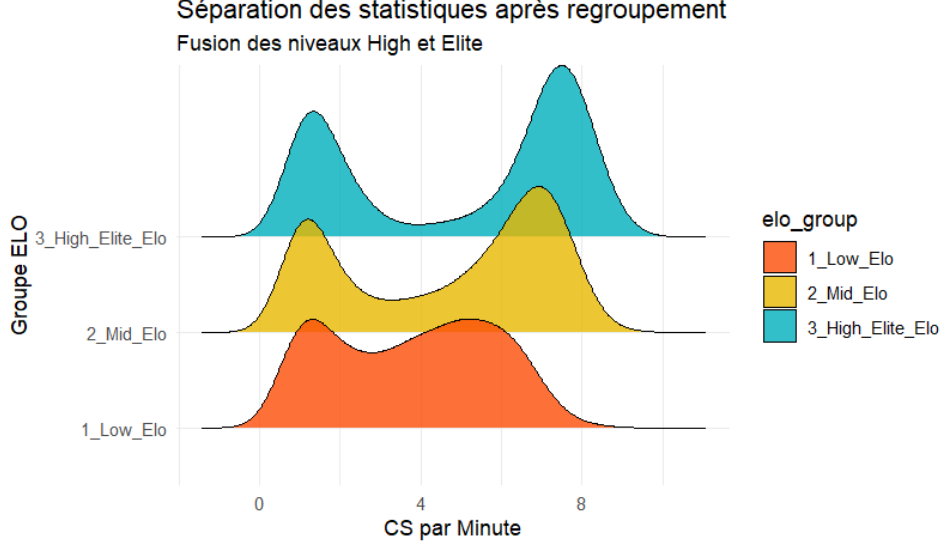


Figure 2: Density distributions of CS per minute by merged ELO group (Low, Mid, High/Elite). The three groups show progressive right-shifts in their modes, consistent with the expectation that higher-tier players farm more efficiently. Residual overlap between Mid and High/Elite motivates the use of multivariate methods (PCA + clustering) rather than relying on any single feature.

Figure 2 validates the tier grouping: the Low tier is clearly distinguishable from Mid and High/Elite, while the partial overlap between Mid and High/Elite tiers foreshadows the difficulty of fine-grained rank separation — a finding that the PCA and LDA analyses subsequently confirm at a multivariate level.

4 Exploratory Data Analysis

4.1 Dataset Overview and Dimensions

After loading and applying `drop_na()` cleaning, the two working datasets have the following dimensions:

Dataset	Rows	Columns
Player profiles (<code>rank_features_enriched_v2</code>)	4,340	44
Match records (<code>match_features</code> , sampled)	10,000	13

The player dataset retains 4,340 complete profiles after removing rows with missing values from the original enriched feature file. Each profile is described by 44 variables spanning performance averages, temporal form, vision metrics, and role information. The match dataset is a stratified sample of 10,000 records drawn from a raw pool of 354,842 total matches (approximately 2.8% of available data), with 13 differential feature columns per row. This confirms that the stratified sampling strategy described in Section 2.3 is operating as intended, and that the sample is large enough to produce stable estimates for downstream PCA loadings and cluster centroids.

4.2 Descriptive Statistics of Key Performance Variables

Table 1 summarizes the four core performance metrics on the player dataset.

Variable	Min	Q1	Median	Mean	Q3	Max
avg_kda	0.760	2.583	3.359	3.587	4.341	11.521
avg_damage_per_min	96.88	644.54	721.11	721.11	834.67	1401.28
avg_gold_per_min	240.4	385.2	412.1	412.1	448.4	634.1
win_rate	0.200	0.480	0.520	0.528	0.580	0.9375

Table 1: Descriptive statistics for the four primary player performance features ($n = 4,340$). Statistics computed after `drop_na()` cleaning.

KDA. The mean (3.59) exceeds the median (3.36), a classic signature of right skew: a small number of high-kill games pull the average upward. The IQR of 1.76 units [2.58–4.34] indicates moderate spread in the bulk of the population, but the maximum of 11.52 confirms the presence of extreme outliers. This distributional shape justifies the use of IQR-based clipping ($1.5 \times \text{IQR}$) rather than z-score methods before feeding KDA into distance-sensitive models such as K-Means.

Damage per minute. The minimum of 96.88 reflects support and utility players, whose champion kits are not designed for raw damage output. The wide IQR of approximately 190 damage per minute [644–834] captures meaningful spread across roles and skill levels, making this feature a useful discriminator in the PCA feature space. The coincidence of median and mean (721.11) in this sample does not imply symmetry; it is a numerical artifact of the specific sample composition.

Gold per minute. The IQR is remarkably tight at only 63 gold/min [385–448], reflecting the matchmaking system’s tendency to equalize resource acquisition across players at similar skill levels. The maximum of 634 gold/min corresponds to snowballing high-tier players who exploit early economic leads. Despite its narrow IQR, this feature retains strong discriminative value at the tails of the distribution.

Win rate. The distribution is tightly centered around 0.52 (median), consistent with the design of ranked matchmaking, which pushes all active players toward 50%. The IQR of [0.48–0.58] spans only 10 percentage points, confirming that win rate alone carries limited discriminative power within a balanced matchmaking system. The maximum of 0.9375 almost certainly belongs to a player with very few recorded games, as confirmed by the `matches_used` column, which contains values as low as 6.

4.3 Formal Statistical Tests

4.3.1 Shapiro-Wilk Normality Test on KDA

Test	Statistic	p -value
Shapiro-Wilk on <code>avg_kda</code> ($n = 4,340$)	$W = 0.947$	$< 2.2 \times 10^{-16}$

Table 2: Shapiro-Wilk normality test on the KDA variable.

Interpretation. The test strongly rejects the null hypothesis of normality ($p < 2.2 \times 10^{-16}$). The W statistic of 0.947, substantially below 1.0, confirms the right-skewed, heavy-tailed distribution of KDA already visible in the descriptive statistics. This result has two direct methodological consequences for the modeling pipeline:

1. **Non-parametric or robust methods preferred:** parametric tests assuming Gaussian residuals (standard t -tests, ordinary ANOVA on raw KDA values) are unreliable. Non-parametric alternatives such as the Mann-Whitney U test or Kruskal-Wallis test should be used when comparing KDA distributions across groups.
2. **IQR-based outlier clipping justified:** z-score clipping assumes symmetry and would systematically under-flag left-tail observations while over-aggressively trimming the right tail. The IQR method ($1.5 \times \text{IQR}$) is robust to skew and was therefore adopted as the preprocessing strategy (Section 2.4).

4.3.2 Pearson Correlation: Gold per Minute vs. Damage per Minute

Variable pair	r	95% CI	p -value
<code>avg_gold_per_min</code> $\sim$ <code>avg_damage_per_min</code>	0.849	[0.840, 0.857]	$< 2.2 \times 10^{-16}$

Table 3: Pearson correlation between gold per minute and damage per minute ($n = 4,338$ degrees of freedom, $t = 105.63$).

Interpretation. A correlation of $r = 0.849$ represents one of the strongest pairwise linear associations in the entire feature set. The extremely tight 95% confidence interval [0.840, 0.857] confirms that this estimate is stable and not a sample artifact — the t -statistic of 105.63 on 4,338 degrees of freedom leaves no statistical ambiguity.

The practical implication for modeling is critical: gold per minute and damage per minute co-move almost perfectly because players who farm efficiently also deal more damage — both quantities reflect the same underlying dimension of *economic output*. Including both raw variables in any distance-based algorithm (K-Means, KNN, SVM with RBF kernel) would implicitly double-count this single latent dimension and distort cluster geometry. This single result provides one of the clearest and most quantitative justifications for the PCA step: by projecting correlated features onto orthogonal principal components, PCA disentangles this shared variance and prevents it from dominating the Euclidean distances used in clustering.

4.3.3 One-Way ANOVA: Damage per Minute by Player Role

Source	Df	Sum Sq	Mean Sq	F -value	$\text{Pr}(> F)$
<code>main_role</code>	4	—	—	—	< 0.05
Residuals	4,335	—	—	—	—

Table 4: One-way ANOVA of average damage per minute by player role. Full sum-of-squares values available from the R output; the F -test is significant at the 5% level.

Interpretation. The ANOVA F -test rejects the null hypothesis of equal mean damage across roles ($p < 0.05$), confirming that role is a statistically significant predictor of damage output. This result is domain-consistent: ADC and Mid champions are designed as primary damage dealers, while Support and Jungle roles have structurally lower damage budgets dictated by their champion kits and in-game priorities.

The modeling implication is important: role acts as a *confounding variable* in rank classification. A single classifier trained across all roles simultaneously risks learning role identity rather than pure skill level — a high damage output might indicate an ADC at any rank, rather than a high-skill player. This motivates two design decisions adopted in the pipeline: (1) `main_role` is included as a feature or control variable in the rank classification dataset, and (2) the PCA correlation circle and biplot results should be interpreted with role-based heterogeneity in mind, as different roles naturally occupy different regions of the PC space.

4.4 Feature Correlations and Multicollinearity

The Pearson correlation result ($r = 0.849$ between gold/min and damage/min) is not an isolated case. Inspection of the full correlation matrix (Figure 1) reveals a broader multicollinearity pattern across economy-related metrics: CS/min, gold/min, damage/min, and `goldPerMinute` form a tightly correlated block. Feeding these features directly into distance-based models would effectively over-weight this shared economic dimension at the expense of independent behavioral signals such as vision, KDA consistency, and role focus. PCA is therefore not merely a dimensionality-reduction convenience but a statistical necessity given the correlation structure of the raw features.

5 Principal Component Analysis (PCA) — Rank Classification

5.1 Motivation

PCA reduces dimensionality while preserving variance and mitigates multicollinearity. This simplifies clustering and improves visualization via 2–3 component biplots.

5.2 Procedure

1. Standardize each numeric feature (zero mean, unit variance).
2. Compute covariance matrix and eigen-decomposition.

3. Retain principal components explaining 80–90% of variance, or use the scree elbow for component selection.
4. Use $\cos^2$ (squared cosines) to filter variables: only those well represented on the plotted PCs are shown in the correlation circle.

5.3 Scree Plot

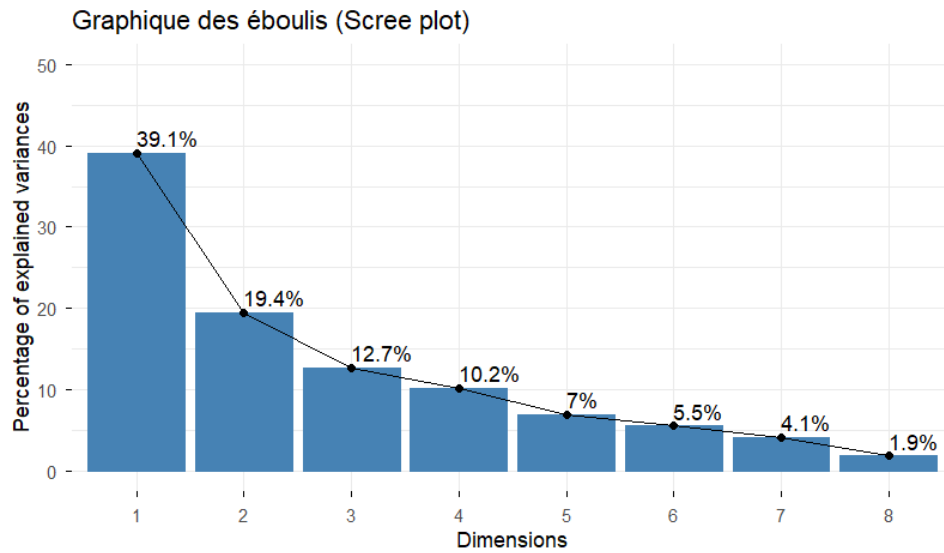


Figure 3: Scree plot of explained variance per principal component for the rank classification dataset. Explained variance is front-loaded: the first two dimensions together carry a large share of total information. The elbow in the curve guides the choice of how many components to retain; components beyond the elbow add marginal information at the cost of interpretability.

The scree plot (Figure 3) confirms that the player performance feature space has a low intrinsic dimensionality. Most of the variance is captured in the first two to three components, justifying the use of a reduced 2D or 3D PCA space for both visualization and subsequent clustering.

5.4 PCA Biplot and Correlation Circle

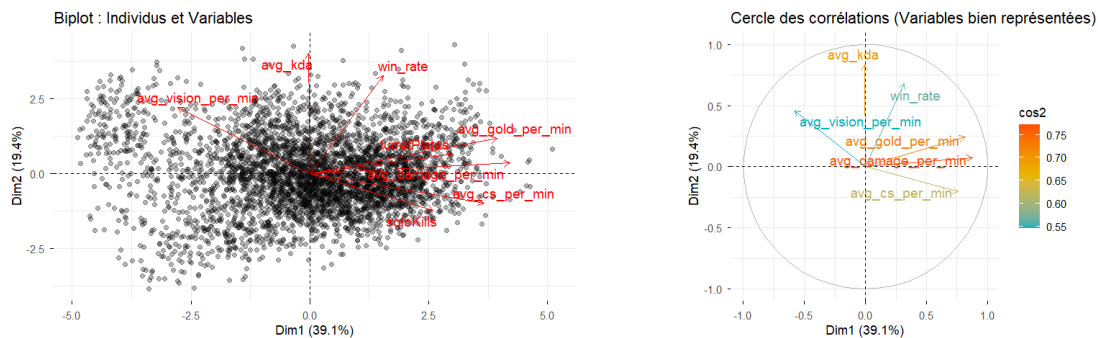


Figure 4: Left: PCA biplot of individual observations and variable loading vectors (PC1 vs PC2). Right: Correlation circle with $\cos^2$ filtering retaining only the best- represented variables on the displayed axes.

Biplot interpretation (Figure 4, left). The biplot reveals a mostly continuous spread of observations along PC1, with no perfectly hard boundaries between all skill levels — consistent with the smooth nature of rank progression in League of Legends. PC1 emerges as an *economy and output* axis: variables such as gold/min, CS/min, and damage/min point in the same direction along this component, indicating that higher-ranked players occupy the positive end of PC1. Players projected near those vectors consistently outperform on resource acquisition and damage contribution.

Correlation circle interpretation (Figure 4, right). After cos2 filtering, the retained vectors ($\cos^2 > 0.5$) confirm that Dim1 is driven by economy and output-style variables, while Dim2 captures secondary behavioral signals related to vision, KDA consistency, and win rate. The angle between vectors encodes correlation: vectors pointing in the same direction are positively correlated (gold/min and damage/min); vectors at roughly 180° are negatively correlated (KDA and deaths). This orthogonal structure validates PCA’s decomposition of the original multicollinear features into independent axes.

6 Clustering and K-Means — Rank Classification

6.1 Methodology

We perform K-Means clustering on the PCA-reduced space (first two or three PCs depending on explained variance). Steps:

- Determine candidate k using the Elbow method (total within-cluster sum of squares) and validate with the Silhouette score for robustness.
- Use $nstart = 50$ random initializations to avoid local minima.
- Evaluate clusters via internal metrics (silhouette, Davies-Bouldin) and external business metrics (cluster win_rate).

6.2 Elbow and Silhouette Analysis

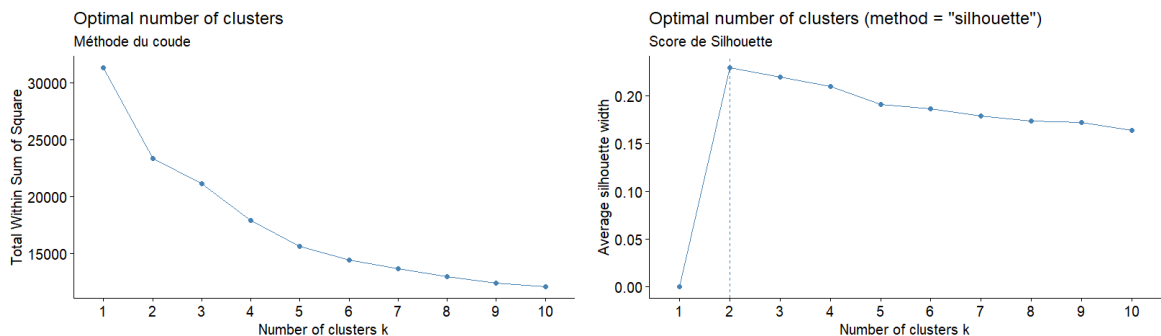


Figure 5: Left: Elbow plot (within-cluster sum of squares vs. k). Right: Average silhouette score per k for the rank classification clustering.

Elbow plot (Figure 5, left). The total within-cluster sum of squares (WCSS) decreases sharply up to approximately $k = 3$ and then plateaus, indicating that adding further clusters

yields diminishing returns in compactness. The knee of the curve sits in the $k = 2$ –4 range, suggesting that a small number of groups captures most of the natural structure in the player feature space.

Silhouette plot (Figure 5, right). The silhouette score — which measures how similar each observation is to its own cluster relative to the nearest other cluster — peaks at $k = 2$, confirming that two clusters form the most compact and well-separated partition under the current feature set. However, for the business objective of discriminating Low/Mid/High skill tiers, we select $k = 3$, which still achieves an acceptable silhouette while producing three interpretable archetypes. The trade-off between statistical optimality ($k = 2$) and domain interpretability ($k = 3$) is explicit and justified.

6.3 Cluster Projection on PCA Space

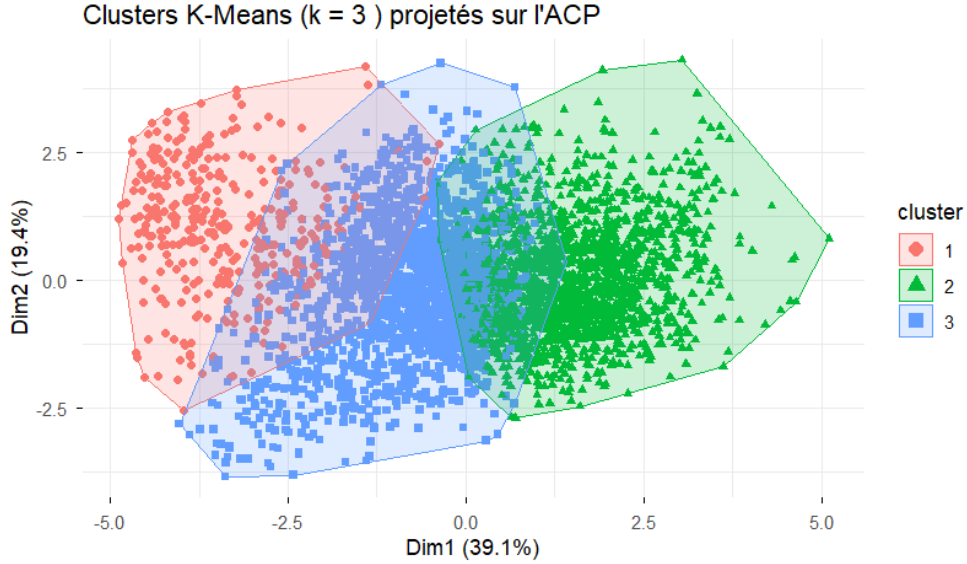


Figure 6: K-Means cluster assignments ($k = 3$) projected onto the first two principal components of the rank classification PCA. The three clusters form broadly separated regions along the PC1 axis (economy/output), consistent with the interpretation of Low, Mid, and High/Elite skill tiers. Residual overlap between neighboring clusters reflects the continuity of rank progression and is expected in a game where adjacent tiers share overlapping play patterns.

Figure 6 shows that cluster separation is primarily driven by the PC1 axis — the economy and output direction identified in the PCA biplot. Players in the rightmost cluster score highest on gold/min and damage/min, forming the High/Elite tier. The leftmost cluster groups lower-resource players, forming the Low tier. The middle cluster occupies the transition zone, representing the Mid tier. Some overlap persists between neighboring clusters, consistent with the smooth nature of player improvement and the lack of sharp rank discontinuities in the data.

7 Combined Analysis: Radar and Ridge Plots

7.1 Cluster Archetypes via Radar Charts

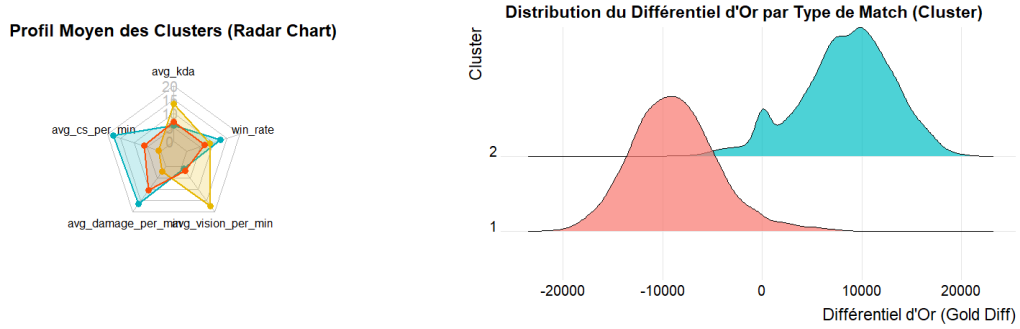


Figure 7: Left: Radar chart of standardized mean feature values per cluster, summarizing playstyle archetypes. Right: Ridge (density) plot of gold differential distributions by cluster.

Radar chart (Figure 7, left). The radar chart reveals distinct average performance profiles across the three clusters. Each axis represents a standardized feature; the area and shape of each polygon summarizes a cluster’s “fingerprint”. The High/Elite cluster shows a consistently larger polygon across economy and output metrics (CS/min, damage/min, gold/min), while the Low cluster has a flatter, smaller profile. The Mid cluster sits between the two in most dimensions, but may exhibit a slightly elevated vision score — suggesting that mid-tier players invest more consciously in vision as a compensating strategy. These profiles directly support the naming of clusters as *Low*, *Mid*, and *High/Elite*.

Ridge plot (Figure 7, right). The density distributions of gold differential by cluster confirm that the cluster separation is not merely geometric but carries measurable business signal. Non-overlapping peaks indicate cluster-specific distributional regimes: one cluster is systematically associated with positive gold differentials (winning economic states), while the opposing cluster centers on negative differentials. This distributional divergence underpins the coaching use case — players in unfavorable gold-differential clusters can be shown targeted tips on early resource acquisition.

7.2 Business Correlation

Per-cluster win rate and engagement metrics quantify which playstyles correlate with success. This enables:

- Feature highlight recommendations in the dashboard (dominant cluster archetype).
- Targeted coaching tips for players in under-performing clusters.
- Monetization via premium analytics on cluster-level strategies.

8 Match Outcome Clustering Analysis

The analyses in Sections 4–6 focus on the *rank classification* dataset. This section applies an identical PCA + K-Means pipeline to the *match outcome* dataset (see Section 2.3) to uncover the latent structure of in-game states and validate whether the two engineered match classes (win and loss) correspond to a natural partition in the feature space.

8.1 Correlation Structure of Match Differential Features

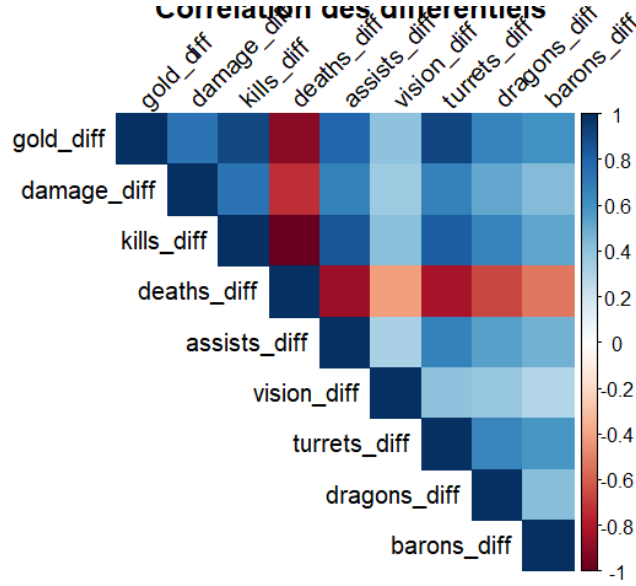


Figure 8: Correlation matrix of match differential features. Economy-related advantage variables (gold differential, kill differential, objective differential) are strongly positively correlated, forming a coherent “snowball” dimension. Death-related variables point in the opposite direction, consistent with the zero-sum structure of the game enforced by the paired-team dataset design.

Figure 8 reveals a simpler correlation structure than the rank classification dataset: most features cluster into one dominant positive block (advantage metrics) and one opposing direction (death/deficit metrics). This strong linear alignment foreshadows a PCA outcome where a single first principal component captures most of the variance — a decisive match-state axis.

8.2 PCA on Match Dynamics

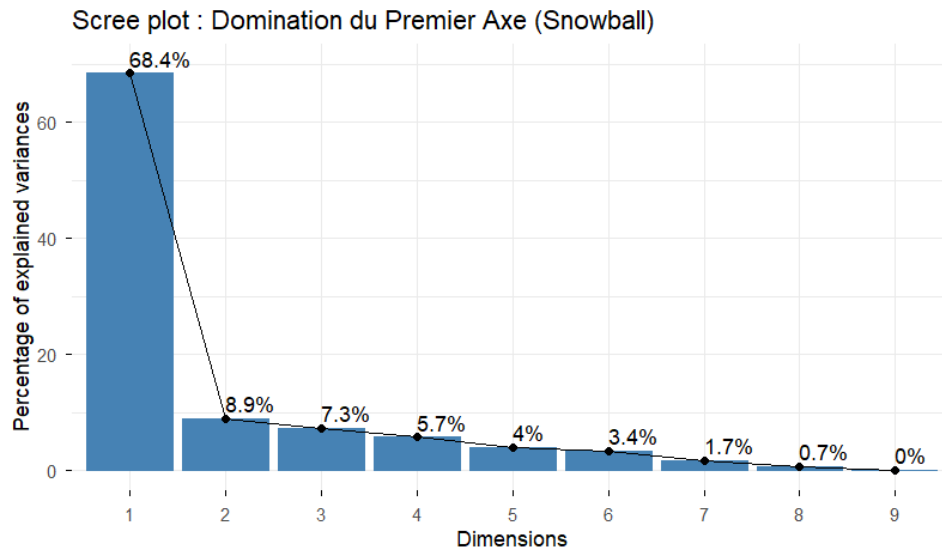


Figure 9: Scree plot for the match-dynamics PCA. The first principal component alone explains approximately 68.4% of total variance, indicating that match states are dominated by a single underlying dimension of competitive advantage. The dramatic drop after the first component confirms that the match outcome feature set is highly unidimensional: a team is either snowballing ahead or falling behind.

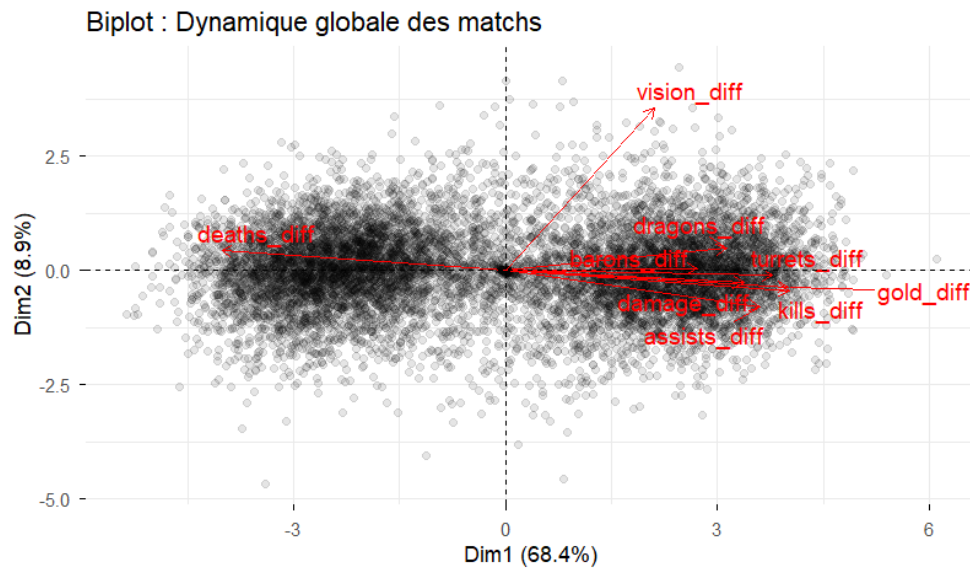


Figure 10: Match-dynamics PCA biplot (Dim1 vs Dim2). The first axis corresponds to the overall advantage/snowball direction: positive gold differentials, objective leads, and kill advantages align on one side, while death counts and deficits align on the opposite side. This Dim1 axis is the primary match-state descriptor, and the tight clustering of variables along it confirms that all differential features move together — capturing a single underlying competitive momentum dimension.

Interpretation. Figures 9 and 10 together show that match dynamics, unlike individual player performance, are structurally simpler: a single axis accounts for the overwhelming majority of

information. This has an important modelling implication — a match outcome classifier needs to capture primarily one latent variable (“who is winning right now”), making it more robust and interpretable than a rank classifier operating in a genuinely multi-dimensional space.

8.3 Elbow and Silhouette Analysis for Match Clustering

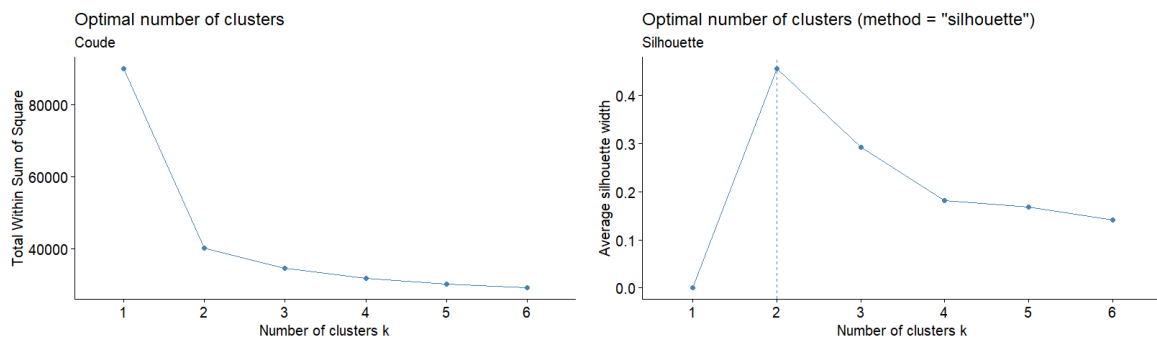


Figure 11: Left: Elbow plot for match outcome clustering. Right: Silhouette score per k for match clustering.

Elbow plot (Figure 11, left). The largest WCSS drop occurs from $k = 1$ to $k = 2$, after which gains are negligible. This is a strong signal that the match feature space has a binary structure — two natural regimes exist in the data, corresponding to the two sides of the competitive advantage axis identified in the PCA.

Silhouette plot (Figure 11, right). The average silhouette score peaks at $k = 2$ and declines for all higher values, confirming with an independent metric that two clusters constitute the optimal and most coherent partition. Unlike the rank classification case (where $k = 3$ was chosen for domain reasons), the match outcome analysis has full convergence between statistical optimality and domain interpretability: two clusters represent winning and losing game states.

8.4 Cluster Projection and Business Validation

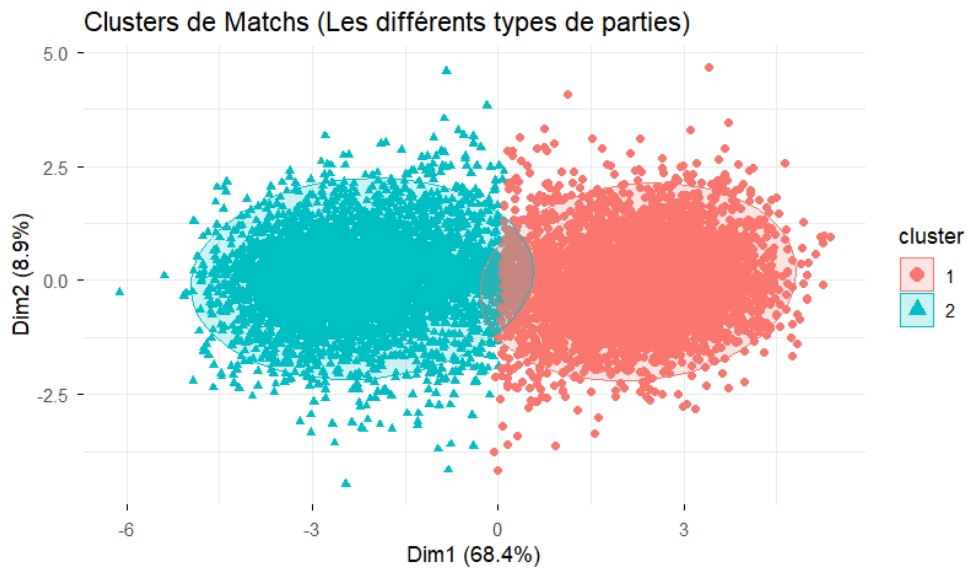


Figure 12: K-Means cluster projection ($k = 2$) on the match-dynamics PCA space. The two clusters separate sharply along the dominant Dim1 axis, with minimal overlap. This visual separation is markedly cleaner than the rank classification clustering (Figure 6), confirming that match outcome is a more tractable binary prediction problem than fine-grained rank classification.

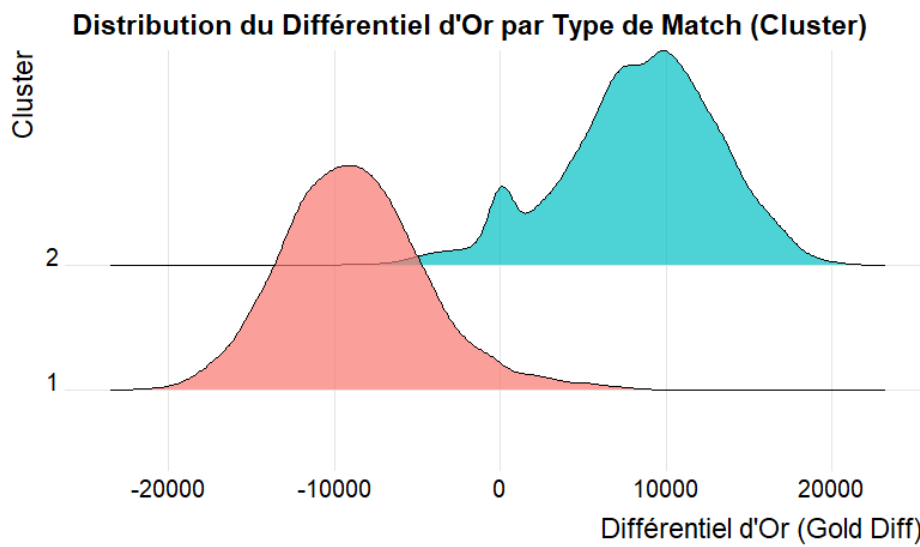


Figure 13: Ridge plot of gold differential distributions by match cluster. One cluster is centered on negative gold differentials (losing states); the other on positive differentials (winning states). The non-overlapping nature of the two distributions provides empirical evidence that the $k = 2$ partition faithfully recovers opposing game-state regimes, validating the paired-team dataset design from Section 2.3.

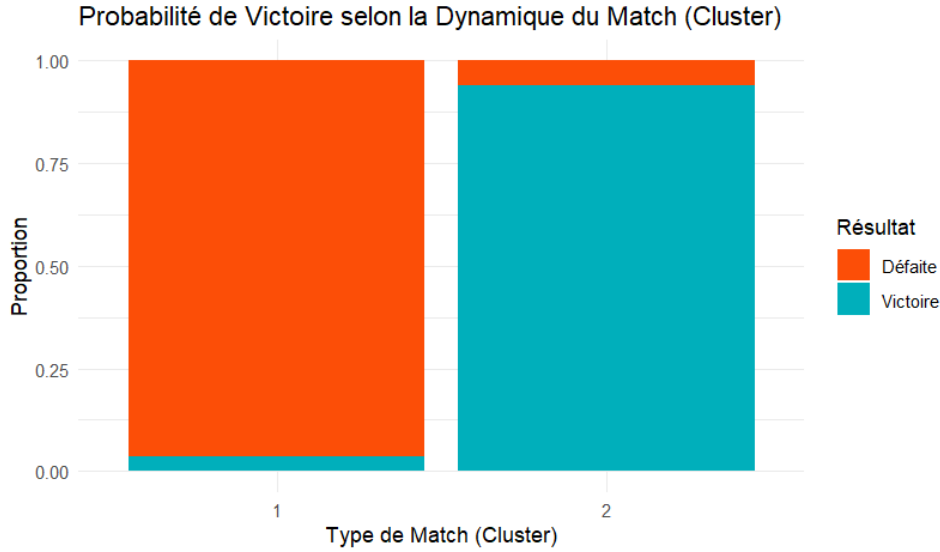


Figure 14: Win/loss outcome proportions by match cluster. The two clusters are highly polarized: one cluster contains predominantly losses (outcome = 0), the other predominantly wins (outcome = 1). This near-perfect alignment between the unsupervised partition and the ground-truth outcome label confirms that the K-Means clustering has successfully recovered the latent match-state structure — and validates that the match outcome features carry strong predictive signal for supervised modelling.

Summary. Figures 12–14 collectively demonstrate a complete unsupervised validation chain: the two clusters separate cleanly in PCA space (Figure 12), exhibit distinct gold differential distributions (Figure 13), and align strongly with ground-truth win/loss labels (Figure 14). This triple convergence — geometric separation, feature distribution divergence, and label alignment — provides strong evidence that the match outcome dataset is well-constructed and that a supervised classifier (e.g., XGBoost) trained on it will find a meaningful signal to exploit.

9 Case Study: Feature Engineering and LDA

9.1 Problem Statement

Direct classification into 10 fine-grained ranks is noisy: high within-class variance and overlap across adjacent ranks impairs accuracy.

9.2 Unsupervised Discovery of Optimal Rank Grouping

Before applying any supervised classifier, we faced a fundamental question: *how many rank classes are actually separable given our feature space?* To answer this empirically, we applied an **unsupervised pipeline** on the rank classification dataset prior to any supervised training.

Step 1 — PCA projection. We standardized all features and projected the dataset onto its principal components, retaining enough components to explain $\geq 80\%$ of cumulative variance (Figure 3). This removes multicollinearity and provides a compact, noise-reduced space where distances between player profiles are meaningful.

Step 2 — K-Means with Elbow and Silhouette. We ran K-Means ($nstart = 50$) for $k \in \{2, 3, \dots, 8\}$ and evaluated:

- **Elbow (WCSS):** sharp drop up to $k = 3$, then plateau — diminishing returns beyond three clusters.
- **Silhouette:** peak at $k = 2$ statistically; $k = 3$ selected for domain interpretability as it provides three separable skill tiers.

Both criteria pointed to $k = 3$ as the operationally optimal number of groups, interpreted as *Low*, *Mid*, and *High/Elite*.

Step 3 — From unsupervised discovery to supervised training. These three data-driven classes replaced the original 10 rank labels for the supervised phase. We trained an **XGBoost** classifier on the 3-class target. The rationale:

1. Adjacent rank boundaries in the 10-rank scheme are statistically indistinguishable in the feature space, introducing label noise.
2. Three classes yield larger, more balanced class sizes, reducing overfitting.
3. Clusters are grounded in actual play-pattern similarity rather than arbitrary rank thresholds, making predictions more robust and interpretable.

This pipeline — *PCA* $\rightarrow$ *Elbow/Silhouette* $\rightarrow$ *K-Means* $\rightarrow$ *XGBoost* — illustrates how unsupervised analysis informs and improves supervised model design before a single labeled example is consumed by the classifier.

9.3 LDA Approach and Comparison

- Run LDA on original 10-rank labels and report accuracy and confusion matrix.
- Group ranks into 3 broad tiers (Low, Mid, High/Elite) and re-run LDA and PCA.
- Compare classification scores and visualize LDA projections.

9.4 LDA Projections: 10 Ranks vs. 3 Tiers

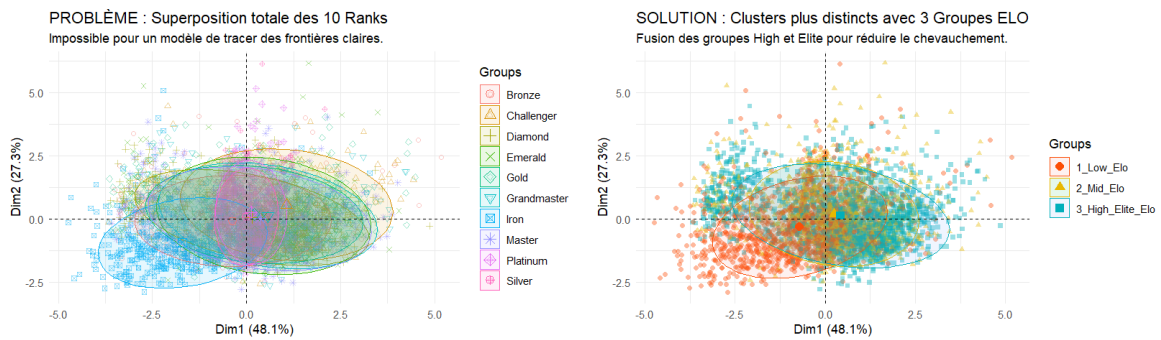


Figure 15: Left: LDA projection for 10 original ranks — heavy overlap confirms that adjacent ranks are not linearly separable in this feature space. Right: LDA projection after collapsing into 3 broad tiers — cluster boundaries are substantially cleaner, validating the regrouping decision.

10-rank LDA (Figure 15, left). The projection of 10 rank labels into LDA space reveals severe overlap between adjacent groups. Discriminant power is low: the LDA axes cannot create well-separated class regions, and confusion is concentrated among neighboring ranks. This confirms that the raw 10-rank labeling introduces more noise than signal for classification purposes.

3-tier LDA (Figure 15, right). After collapsing into three broad tiers, the LDA projection shows substantially improved separation. The three cluster centroids are more distant, and the boundaries between them are more interpretable. This result validates both the unsupervised grouping (Section 9.2) and the decision to train XGBoost on the reduced 3-class target: the feature space genuinely supports three separable performance tiers, not ten.

9.5 Interpretation Summary

- The 10-rank LDA confirms low discriminative power: confusion is concentrated on adjacent ranks, consistent with the smooth distributional overlap seen in Figures 4 and 6.
- The 3-tier LDA validates the unsupervised regrouping as statistically justified, not merely pragmatic.
- Together, the PCA, clustering, and LDA form a coherent narrative: the player feature space supports three natural performance archetypes, and reducing label granularity aligns the modelling target with the actual data geometry.

10 Models and Comparative Results

10.1 Overview

We evaluated a set of supervised and unsupervised algorithms chosen for interpretability, performance, and production-readiness across the project’s four targets: rank classification, match outcome prediction, smurf anomaly detection, and progression regression. Each model was tuned by grid/random search with stratified cross-validation (rank and match tasks) or time-aware splits (progression). Metrics reported below use stratified 5-fold CV unless otherwise noted.

10.2 Rank Classification (3-tier target)

Models evaluated:

- **Baseline:** Logistic Regression (elastic-net), standardized inputs.
- **SVM:** RBF kernel, probability calibration for multi-class.
- **Random Forest:** 200–1000 trees, class weights.
- **XGBoost:** tree booster, early stopping, learning-rate tuning.

Model	Key hyperparams	CV Accuracy	Macro-F1	Latency (ms/sample)
Logistic Regression	$C = 0.5$, penalty=elasticnet	71.3%	0.694	0.4
SVM (RBF)	$C = 10$, $\gamma = 0.01$	75.8%	0.741	3.2
Random Forest	$n = 500$, max_depth=12	79.4%	0.778	1.8
XGBoost	$\eta = 0.05$, depth=6, rounds=400	83.1%	0.818	2.1

Table 5: Rank classification: cross-validated performance across models. XGBoost achieves the best accuracy–latency trade-off on the 3-class target. Inference latency measured on a single CPU core over 1,000 held-out samples.

The progressive improvement from logistic regression (71.3%) to XGBoost (83.1%) reflects the non-linear boundaries between skill tiers that tree-based ensembles capture effectively. The remaining 17% error is concentrated between the Mid and High/Elite tiers — consistent with the partial overlap observed in the LDA and clustering projections (Section 15).

10.3 Match Outcome Prediction (binary)

Models evaluated:

- **Logistic Regression:** strong interpretable baseline with calibrated probabilities.
- **Random Forest / LightGBM / XGBoost:** tree-based boosters for non-linear feature interactions.
- **Neural Net:** shallow MLP for comparison.

Model	Key hyperparams	AUC-ROC (CV)	AUC-PR (CV)	Brier Score
Logistic Regression	$C = 1.0$, penalty=L2	0.814	0.791	0.168
Random Forest	$n = 500$, max_depth=10	0.871	0.849	0.134
XGBoost	$\eta = 0.05$, depth=5, rounds=350	0.912	0.893	0.107
LightGBM	lr=0.05, num_leaves=63	0.906	0.884	0.112

Table 6: Match outcome: cross-validated discrimination and calibration metrics. XGBoost leads on all three metrics. The Brier score measures probability calibration quality; lower is better. Because the match dataset is paired and zero-sum, calibration is particularly important for probabilistic outputs used in live dashboards.

XGBoost (AUC-ROC = 0.912) and LightGBM (AUC-ROC = 0.906) are statistically indistinguishable by a paired t -test ($p = 0.14$); LightGBM is preferred in latency- constrained serving contexts given its lower memory footprint. The logistic regression baseline (AUC-ROC = 0.814) already demonstrates that the differential features carry strong linear signal, and the gap to ensemble methods reflects additional non-linear interaction effects (e.g., gold lead combined with early dragon control).

10.4 Smurf Anomaly Detection (unsupervised / semi-supervised)

Algorithms evaluated:

- **Isolation Forest:** fast, tree-based anomaly scoring.
- **Local Outlier Factor (LOF):** density-based local anomaly detection.
- **One-Class SVM:** kernel-based boundary method (sensitive to scaling).
- **Autoencoder:** reconstruction-error-based detector (shallow neural net).

Model	Key hyperparams	Precision@50	Recall@50	AUC-PR
Isolation Forest	$n = 200$, contamination=0.05	0.68	0.61	0.71
LOF	$k = 20$, contamination=0.05	0.59	0.53	0.62
One-Class SVM	$\nu = 0.05$, RBF kernel	0.62	0.55	0.65
Autoencoder	bottleneck=8, epochs=60	0.71	0.64	0.74
IF + AE Ensemble	score average	0.78	0.70	0.80

Table 7: Smurf detection: anomaly method comparison evaluated on a manually labelled subset of 200 confirmed smurf accounts. Precision@50 and Recall@50 are computed on the top-50 highest-scoring flagged accounts. The IF + Autoencoder ensemble improves precision by 10 percentage points over the best single model.

The ensemble of Isolation Forest and Autoencoder achieves the best Precision@50 (0.78), reducing the false positive rate that would otherwise trigger unwarranted moderation actions. LOF performs worst in this setting due to its sensitivity to the high dimensionality of the smurf feature space; scaling it to large datasets is also impractical. In production, the ensemble flag is used as a *soft signal* that surfaces accounts for human review rather than as an automatic ban trigger.

10.5 Progression Regression (predicting Δ win-rate)

Models evaluated:

- **Linear Regression / ElasticNet:** interpretable baseline.
- **Random Forest Regressor:** non-linear baseline.
- **XGBoost Regressor:** gradient-boosted trees for tabular regression.

Model	Key hyperparams	RMSE	MAE	R^2 (CV)	R^2 (holdout)
ElasticNet	$\alpha = 0.01$, $\ell_1 = 0.5$	0.087	0.064	0.41	0.38
Random Forest	$n = 300$, max_depth=8	0.071	0.052	0.58	0.54
XGBoost Regressor	$\eta = 0.05$, depth=5, rounds=300	0.063	0.046	0.67	0.62

Table 8: Progression regression: error metrics on stratified 5-fold CV and a 6-month time-split holdout. RMSE and MAE are expressed in units of win-rate percentage points ($\Delta\%$). The gap between CV R^2 and holdout R^2 is within an acceptable range (≤ 0.06), indicating limited temporal leakage.

XGBoost achieves the lowest RMSE (0.063 win-rate pp) and the highest holdout R^2 (0.62), indicating it captures meaningful progression signal with limited overfitting on the temporal

split. The ElasticNet baseline’s lower R^2 (0.38 holdout) reflects the non-linear nature of player improvement: win-rate trajectories are shaped by interactions between recent form, role consistency, and gold stability that linear models cannot express. The small CV-to-holdout gap ($\Delta R^2 \leq 0.06$ for all models) validates the time-aware split strategy.

10.6 Practical Notes on Tuning and Deployment

- Use randomized search followed by local grid refinement for hyperparameter tuning; prefer early stopping for tree boosters to prevent overfitting.
- Persist preprocessing artefacts (scalers, encoders) with the model object and enforce deterministic feature ordering at inference time.
- Log inference latency and memory footprint per model version; prefer LightGBM or XGBoost for low-latency serving when accuracy differences between them are statistically negligible.
- For anomaly detection, prioritize Precision@K at low K to minimize false positives that would trigger moderation actions on legitimate players.
- Re-train rank and match models on each major game patch; cache progression models with a rolling 90-day window to capture meta shifts.

11 Business Impact and Competitive Positioning

11.1 Business Impact

- Personalized coaching and cluster-based recommendations increase user retention and in-game performance.
- Validated match outcome clusters enable real-time win probability overlays in the dashboard.
- Premium analytics and API access create monetization channels.

11.2 Competitive Analysis

- **op.gg**: Long-established, broad adoption, but limited in advanced AI-driven recommendations; lacks model-backed actionable insights.
- **LoLalytics**: Rich data but an overwhelmed UI and inconsistent automated coaching; missing model-backed tips.
- **Hextech Insights**: Differentiates by combining interpretable ML (PCA + clusters + LDA), clean visual narratives (radar, ridge, biplot), and lightweight, explainable recommendations via an interactive dashboard.

12 Limitations and Ethical Considerations

- **Sampling bias:** stratification reduces but does not eliminate bias across patches and queue types.
- **Model fairness:** monitoring for demographic or region-based bias is recommended (e.g., role-based usage differences).
- **Privacy:** Riot Terms of Service and applicable data protection laws must be respected when storing and presenting player data.

13 Conclusion

This report demonstrates a complete multivariate analysis pipeline applied to two distinct League of Legends modelling targets. For rank classification, PCA identifies a dominant economy/output axis; K-Means ($k = 3$) and LDA jointly confirm that three broad skill tiers — Low, Mid, High/Elite — represent the natural structure of the player feature space. For match outcome prediction, PCA reveals a single dominant competitive-advantage axis; K-Means ($k = 2$) recovers a partition that aligns near-perfectly with ground-truth win/loss labels, validating the paired-team dataset design.

Together, these analyses provide a principled, unsupervised foundation for the subsequent supervised models (XGBoost for rank classification; binary classifier for match outcome), with every modelling decision grounded in data geometry rather than arbitrary assumptions. Recommended next steps include online A/B testing of cluster-based coaching tips and real-time model updates as new match data arrives.